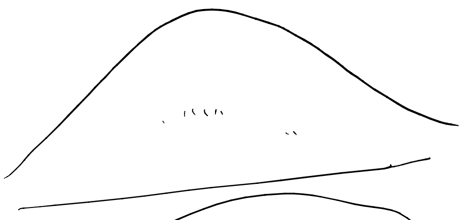
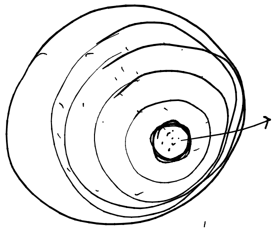
PRST (07/09/2021)

Company – C (system).

Model number – M

X – duraration of such system X –





$x_1 = 10.5$
$x_2 = 10$
$\vdots$
$x_9 = 11$
$x_{10} = 13.5$

$\frac{1}{n}\sum x_i f_i = \sum x_i \left(\frac{f_i}{n}\right)$

## Expectation

(i) X is discrete random variable which can take values $S = \{x_1, x_2, \dots\}$

$$\mu = E(X) = \sum_{i=1}^{\infty} x_i P(X = x_i).$$

(ii) X is cont rv

$$\mu = E(X) = \int_{-\infty}^{\infty} x f(x)\, dx.$$

Variance –

(i) X is discrete with suppor $S = \{x_1, x_2, \dots\}$

$$Var(X) = E(X-\mu)^2.$$
$$= \sum_{i=1}^{\infty} (x_i - \mu)^2 P(X = x_i)$$

(ii) X is cont rv $\quad Var(X) = E(X-\mu)^2$

$$= \int_{-\infty}^{\infty} (x-\mu)^2 f(x)\, dx$$

if X discrete

$$E(g(X)) = \begin{cases} \sum_{i=1}^{\infty} g(x_i)\, P(X=x_i) & \text{if } X \text{ discrete} \\ \int_{-\infty}^{\infty} g(x) f(x)\, dx & \text{if } X \text{ cont.} \end{cases}$$

Eg $X$ is cont rv; pdf $f(x) = \begin{cases} cx^2 & |x| \leq 1 \\ 0 & \text{o.w} \end{cases}$

(i) find $c$

(ii) $E(X)=?$, $Var(X)=?$

Sol: $\int_{-\infty}^{\infty} f(x)\,dx = 1 \Rightarrow \int_{-1}^{1} cx^2\,dx = 1$

$\Rightarrow c = 3/2$

(ii) $\mu = E(X) = \int_{-1}^{1} x f(x)\,dx = \int_{-1}^{1} x\, cx^2\,dx = c \cdot \int_{-1}^{1} x^3\,dx = 0$

$Var(X) = E(X-\mu)^2$

$= E(X^2)$

$= \int_{-1}^{1} x^2 \cdot cx^2\,dx = \frac{3}{2}\int_{-1}^{1} x^4\,dx$

$= \frac{3}{5}$

$$\boxed{E(X-\mu)^2 = E(X^2) - \mu^2}$$

r-th moment about any poin A : $\mu_r' = E[(X-A)^r]$

r-th raw moment : $\mu_r' = E[(X)^r]$

$\mu_r = E[(X-\mu)^r]$

. . . . hod moment

r-th raw moment $\mu_r' = E(\cdot)$

r-th central moment

$$\mu_2 = \mu_2' - \mu_1'^2$$

$$\mu_3 = \mu_3' - 3\mu_2'\mu_1' + 2\mu_1'^3$$

$$\mu_4 = \mu_4' - 4\mu_3'\mu_1' + 6\mu_2'\mu_1'^2 - 3\mu_1'^4$$

$$E[(X-\mu)^r] = E[(X - A + A - \mu)^r]$$

## properties of expectation

① $E(X_1 + X_2 + \cdots + X_k) = E(X_1) + E(X_2) + \cdots + E(X_k)$.

② $E(a + bX) = a + bE(X)$.

③ $Var(a+bX) = b^2 Var(X)$.

$$E(X-\mu)^2$$
$$= E[X^2 + \mu^2 - 2X\mu]$$
$$= E(X^2) + E(\mu^2) - 2E(X\mu).$$
$$= E(X^2) + \mu^2 - 2\mu^2$$
$$= E(X^2) - \mu^2$$

$$E(c) = c$$
$$= \int_{-\infty}^{\infty} c f(x)\,dx$$
$$= c$$

## Moment generating function (MGF)

$X \longrightarrow$ Mgf (at t) $M(t) = E(e^{tX})$

$$e^{tX} = \sum_{r=0}^{\infty} \frac{t^r}{r!} x^r$$

$$E(e^{tX}) = E\left(\sum^{\infty} \frac{t^r}{r!} X^r\right)$$

$$E(e^{tX}) = E\left(\sum_{r=0}^{\infty} \frac{t^r}{r!} X^r\right)$$

$$= \sum_{r=0}^{\infty} \frac{t^r}{r!} \mu_r'$$

$$\frac{d\,M(t)}{dt} = \underline{E(X e^{tX})}$$

$$\left.\frac{dM(t)}{dt}\right|_{t=0} = \underline{E(X)} = \mu$$

$$\frac{d^2M(t)}{dt^2} = E(X^2 e^{tX})$$

$$\left.\frac{d^2M(t)}{dt^2}\right|_{t=0} = E(X^2) = \mu_2'$$

$$\frac{d^r M(t)}{dt^r} = E(X^r) = \mu_r'$$

$q = 1 - p$

Eg $M(t) = (q + pe^t)^n$

$E(X)$, $Var(X) = ?$

$$\frac{d\,M(t)}{dt} = n(q + pe^t)^{n-1} \cdot pe^t$$

$$E(X) = \left.\frac{dM(t)}{dt}\right|_{t=0} = n \times p$$

$n-2$

$$E(X) = \frac{\quad}{dt}\Big/_{t=0}$$

$$E(X^2) = \frac{d^2M(t)}{dt^2}\Big/_{t=0} = np^2(n-1)(q+pe^t)^{n-2}\Big/_{t=0} + np$$

$$= n(n-1)p^2 + np$$

$$Var(X) = E(X^2) - \mu^2$$

$$= n(n-1)p^2 + np - n^2p^2$$

$$= np(1-p).$$

Eg pmf $f(x) = \frac{e^{-\theta}\theta^x}{x!}$ $\quad x = 0, 1, 2, \ldots\ldots$ $\quad p(X=x_i)$

Mgf = ?

$$M(t) = e^{(\theta(e^t-1))}.$$

$$E(e^{tX}) = \sum_{x=0}^{\infty} \frac{e^{tx}e^{-\theta}\theta^x}{x!}$$

$$= e^{-\theta}\sum_{x=0}^{\infty} \frac{(e^t\theta)^x}{x!}$$

$$= e^{-\theta}e^{\theta e^t}$$

$$= e^{\theta(e^t-1)}.$$